

E-Commerce Sales Performance and Customer Insights Dashboard

1. Project Overview and Objective

To analyze e-commerce sales data and develop interactive Power BI dashboards that provide actionable insights into sales performance, customer behavior, product and category performance, and regional sales trends. The project aims to transform raw data into meaningful visualizations, enabling stakeholders to monitor key performance indicators (KPIs), identify trends and opportunities, and support data-driven business decision-making.

2. Data Sources

- **Source Description and Timeline:** Google Dataset Search and 2020-2024.
 - **Domain:** E-commerce / Sales Analytics
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3. Problem Statement

- To analyze overall sales performance across products, categories, and regions to identify revenue trends and business growth opportunities.
 - To study customer purchasing behavior and buying patterns for improved customer engagement and retention.
 - To evaluate product and category performance to identify best-selling and underperforming products.
 - To monitor key business metrics through interactive dashboards for effective performance tracking.
 - To identify regional sales trends and market opportunities to support strategic business decisions.
 - To transform raw e-commerce sales data into meaningful visualizations that enable data-driven decision-making.
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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Customer ID	Integer / String	Unique identifier for each Customer
Customer Name	String (Text)	Name of the customer
City	String (Text)	City where the customer resides
State	String (Text)	State where the customer resides
Country	String (Text)	Country where the customer resides
Order ID	String (Text)	Unique identifier for each order
Order Date	Date	Date when the order was placed
Product ID	Integer / String	Unique identifier for each product
Product Name	String (Text)	Name of the Product
Category	Categorical	Category of the Product
Brand	String (Text)	Brand of the Product
Quantity	Numeric (Integer)	Number of units ordered
Unit Price	Numeric (Decimal)	Price per unit of the product
Discount	Numeric (Decimal)	Discount applied to the order
Tax	Numeric (Decimal)	Tax charged on the product
Shipping Cost	Numeric (Decimal)	Shipping cost of the order
Total Amount	Numeric (Decimal)	Total amount paid for the order
Payment Method	Categorical	Method used for Payment
Order Status	Categorical	Current status of the order
Seller ID	Integer / String	Unique identifier for the seller

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Standardized column names:** Column names were renamed to improve readability and ensure a consistent naming convention (e.g., Customer ID, Customer Name).
- **Standardized column formats:** Column formats were updated based on the nature of the data to ensure proper representation and consistency. Date values were formatted using the Date format, monetary fields such as Unit Price, Tax, Shipping Cost, and Total Amount were formatted as Currency, and percentage-based fields such as Discount were formatted as Percentage.
- **Removed duplicate records:** A duplicate record was identified in the Order Details sheet using Conditional Formatting and removed using the Data → Remove Duplicates feature. .
- **Handled missing Product IDs:** Missing values in the Product ID column were identified and replaced using a combination of the IF, XLOOKUP, and FILTER functions to retrieve the correct Product ID based on existing records.
`=IF(ISBLANK([@[Product ID]]),XLOOKUP([@[Product Name]],FILTER(D:D,C:C<>""),FILTER(C:C,C:C<>"")),[@[Product ID]])`
- **Handled missing Product Names:** Missing values in the Product Name column were identified and replaced using a combination of the IF, XLOOKUP, and FILTER functions to retrieve the correct product name based on existing records.
`=IF(ISBLANK([@[Product Name]]),XLOOKUP([@[Product ID]],FILTER(C:C,D:D<>""),FILTER(D:D,D:D<>"")),[@[Product Name]])`
- **Corrected Product Category values:** The Product Category column contained incorrect values. To resolve this, the unique Product Name and Product Category values were extracted using the UNIQUE() function and organized into a separate worksheet named Product Category. The XLOOKUP() function was then used to retrieve and update the correct category for each product.
- **Standardized Customer Names:** The Customer Name column in the Customer Details table contained inconsistent letter casing and extra spaces. These issues were corrected using a combination of the PROPER() and TRIM() functions to standardize the formatting.

- **Handled missing State values:** Missing values in the State column were identified and replaced using a combination of the IF, XLOOKUP, and FILTER functions to retrieve the correct state based on existing records.

=IF([@State]="",XLOOKUP([@City],FILTER(C:C,D:D<>""),FILTER(D:D,D:D<>"")),[@State])

- The Country column contained inconsistent values (e.g., United States, United States, US, and us). These inconsistencies were identified using the Filter feature and corrected using the Find and Replace tool to ensure a consistent country name throughout the dataset.
- Filtered the dataset to display only pending orders using Excel filters.

	Order Date	Customer ID	Product ID	Product Name	Category	Brand	Quantity	Unit Price	Discount	Tax	Shipping Cost	Total Amount	Payment Method	Order Status	Seller ID
9	10-11-2022	CUST031165	P00028	Jeans	Clothing	KiddoFun	2	\$306.51	5%	\$29.12	\$6.24	\$617.73	Debit Card	Pending	SELL00510
27	08-12-2023	CUST035043	P00030	Dress Shirt	Clothing	Apex	4	\$380.23	20%	\$60.84	\$9.28	\$1,286.86	Amazon Pay	Pending	SELL00342
57	08-03-2024	CUST030091	P00044	Power Bank 20000	Electronics	Apex	4	\$364.08	25%	\$131.07	\$12.95	\$1,236.26	Credit Card	Pending	SELL01113
91	03-02-2023	CUST039077	P00015	Instant Pot	Home & Kitch	ReadMor	5	\$342.26	20%	\$109.52	\$7.29	\$1,485.85	Credit Card	Pending	SELL00271
108	13-10-2022	CUST002237	P00014	Smart Light Bulb	Home & Kitch	UrbanStyl	3	\$297.26	5%	\$101.66	\$5.24	\$954.09	Credit Card	Pending	SELL00371
143	18-08-2020	CUST027712	P00017	Electric Kettle	Home & Kitch	UrbanStyl	5	\$544.52	25%	\$245.03	\$2.79	\$2,289.77	Cash on Delivery	Pending	SELL00796
159	14-11-2021	CUST016025	P00045	Phone Tripod	Electronics	NexPro	2	\$400.69	10%	\$57.70	\$10.68	\$789.62	Credit Card	Pending	SELL01573
187	17-10-2022	CUST021204	P00042	Projector Mini	Electronics	UrbanStyl	2	\$495.32	0%	\$49.53	\$10.01	\$1,050.18	Credit Card	Pending	SELL01469
191	06-07-2023	CUST031415	P00033	Puzzle 1000pc	Toys & Games	UrbanStyl	5	\$493.81	30%	\$138.27	\$14.70	\$1,881.30	Amazon Pay	Pending	SELL00905
193	15-07-2023	CUST019542	P00014	Drone Mini	Electronics	Apex	5	\$274.55	30%	\$48.05	\$5.27	\$1,014.24	Debit Card	Pending	SELL00946
196	11-11-2023	CUST004652	P00028	Jeans	Clothing	ReadMor	5	\$259.57	5%	\$0.00	\$7.79	\$1,240.75	UPI	Pending	SELL00205
199	16-07-2020	CUST007148	P00010	Smartwatch	Electronics	FitLife	1	\$402.92	15%	\$27.40	\$11.53	\$381.41	Net Banking	Pending	SELL00782
199	17-03-2023	CUST039236	P00042	Projector Mini	Electronics	BrightLux	3	\$582.15	0%	\$87.32	\$6.71	\$1,840.48	Debit Card	Pending	SELL01738
145	11-10-2022	CUST041438	P00025	Running Shoes	Sports & Out	Zenith	5	\$39.90	10%	\$8.98	\$9.41	\$197.94	Amazon Pay	Pending	SELL01014
160	06-07-2024	CUST034300	P00038	External HDD 2TB	Electronics	FitLife	2	\$569.77	5%	\$86.61	\$9.56	\$1,178.73	Debit Card	Pending	SELL00010
162	16-12-2023	CUST005859	P00048	Wireless Charger	Electronics	ReadMor	5	\$131.34	5%	\$49.91	\$1.64	\$675.42	Amazon Pay	Pending	SELL01019
120	20-10-2021	CUST037096	P00013	Action Camera	Electronics	Zenith	3	\$495.51	30%	\$124.87	\$4.78	\$1,170.22	Net Banking	Pending	SELL01193
108	28-07-2021	CUST044182	P00039	Graphic Tablet	Electronics	ReadMor	1	\$345.37	5%	\$26.25	\$7.68	\$362.03	Debit Card	Pending	SELL01902
126	12-12-2022	CUST046117	P00009	Portable SSD 1TB	Electronics	HomeEas	4	\$562.49	5%	\$106.87	\$10.79	\$2,255.12	Debit Card	Pending	SELL00235
135	18-09-2022	CUST023759	P00018	Vacuum Cleaner	Home & Kitch	Zenith	4	\$117.96	5%	\$22.41	\$12.41	\$483.07	Debit Card	Pending	SELL00013
154	17-12-2022	CUST031827	P00036	Smart Light Bulb	Home & Kitch	NexPro	4	\$290.45	0%	\$139.42	\$13.20	\$1,214.42	Net Banking	Pending	SELL00965
104	05-08-2020	CUST032624	P00033	Puzzle 1000pc	Toys & Games	ReadMor	2	\$147.61	10%	\$21.26	\$10.40	\$207.36	Debit Card	Pending	SELL01280
107	06-12-2020	CUST012756	P00011	Fitness Band	Sports & Out	NexPro	2	\$433.81	0%	\$69.41	\$1.25	\$938.28	Debit Card	Pending	SELL00695

- Sorted the dataset by Unit Price in ascending order (lowest to highest).

	Order Date	Customer ID	Product ID	Product Name	Category	Brand	Quantity	Unit Price	Discount	Tax	Shipping Cost	Total Amount	Payment Method	Order Status	Seller ID
2	12-03-2023	CUST036170	P00003	Smartphone Case	Electronics	HomeEas	2	\$5.00	0%	\$0.00	\$2.82	\$12.82	Credit Card	Shipped	SELL00062
3	25-12-2024	CUST009505	P00021	Backpack	Sports & Out	Zenith	2	\$5.00	0%	\$0.50	\$6.27	\$16.77	Credit Card	Returned	SELL00560
4	23-09-2023	CUST049762	P00011	Fitness Band	Sports & Out	FitLife	4	\$5.01	15%	\$1.36	\$2.67	\$21.06	Debit Card	Delivered	SELL00756
5	23-01-2020	CUST042993	P00023	Cookware Set	Home & Kitch	KiddoFun	1	\$5.01	15%	\$0.77	\$5.98	\$11.01	Credit Card	Delivered	SELL01462
6	01-08-2022	CUST021018	P00001	Wireless Earbuds	Electronics	ReadMor	3	\$5.02	0%	\$1.20	\$4.74	\$21.00	UPI	Delivered	SELL00772
7	26-11-2024	CUST040824	P00034	Desk Organizer	Home & Kitch	UrbanStyl	5	\$5.02	0%	\$1.25	\$2.46	\$28.81	Credit Card	Delivered	SELL01433
8	08-04-2021	CUST002687	P00024	Yoga Mat	Sports & Out	NexPro	3	\$5.03	10%	\$1.09	\$5.53	\$20.20	Credit Card	Delivered	SELL00053
9	12-08-2020	CUST031990	P00039	Graphic Tablet	Electronics	HomeEas	4	\$5.03	20%	\$0.80	\$1.54	\$18.44	UPI	Cancelled	SELL01294
10	03-10-2020	CUST039988	P00011	Fitness Band	Sports & Out	FitLife	5	\$5.06	20%	\$3.64	\$4.05	\$27.93	UPI	Delivered	SELL00333
11	10-07-2023	CUST015293	P00022	Water Bottle	Sports & Out	Zenith	4	\$5.06	0%	\$1.01	\$7.24	\$28.49	Credit Card	Delivered	SELL01775
12	29-06-2020	CUST027251	P00025	Running Shoes	Sports & Out	KiddoFun	2	\$5.06	5%	\$1.15	\$3.64	\$14.40	Net Banking	Cancelled	SELL00093
13	28-02-2021	CUST044013	P00040	Microphone	Electronics	Apex	2	\$5.07	30%	\$0.35	\$7.92	\$15.37	Amazon Pay	Delivered	SELL00575
14	03-10-2022	CUST022687	P00025	Running Shoes	Sports & Out	CoreTech	5	\$5.08	25%	\$0.95	\$1.33	\$21.33	Credit Card	Shipped	SELL00693
15	04-04-2024	CUST032283	P00034	Desk Organizer	Home & Kitch	ReadMor	2	\$5.10	0%	\$0.51	\$1.69	\$12.40	Net Banking	Delivered	SELL00289
16	06-03-2022	CUST042146	P00008	4K Monitor	Electronics	NexPro	2	\$5.10	25%	\$0.92	\$5.10	\$13.67	Credit Card	Delivered	SELL00678
17	06-04-2024	CUST041684	P00042	Projector Mini	Electronics	UrbanStyl	4	\$5.12	0%	\$1.02	\$6.93	\$28.43	Amazon Pay	Delivered	SELL00122
18	28-04-2024	CUST037432	P00035	Desk Plant	Home & Kitch	KiddoFun	1	\$5.13	20%	\$0.00	\$5.74	\$9.84	Amazon Pay	Cancelled	SELL00718
19	28-04-2023	CUST046435	P00022	Water Bottle	Sports & Out	UrbanStyl	4	\$5.14	0%	\$1.03	\$1.90	\$23.49	UPI	Delivered	SELL00812
20	13-12-2022	CUST001080	P00041	Webcam Full HD	Electronics	BrightLux	3	\$5.14	15%	\$1.57	\$6.32	\$21.00	UPI	Delivered	SELL00104
21	11-10-2023	CUST017426	P00047	Memory Card 128G	Electronics	ReadMor	2	\$5.14	0%	\$0.82	\$4.57	\$15.67	UPI	Delivered	SELL01760
22	11-12-2022	CUST048252	P00007	Mechanical Keyboard	Electronics	HomeEas	3	\$5.15	10%	\$1.67	\$4.44	\$20.02	Debit Card	Delivered	SELL00942
23	25-03-2024	CUST019630	P00033	Puzzle 1000pc	Toys & Games	ReadMor	3	\$5.15	0%	\$0.77	\$6.76	\$22.98	Credit Card	Delivered	SELL01262
24	18-01-2023	CUST028504	P00048	Wireless Charger	Electronics	Zenith	1	\$5.15	0%	\$0.93	\$3.83	\$9.91	Net Banking	Delivered	SELL00891

- Pivot tables were created to analyze sales performance over the past five years, identify the best-performing and underperforming products, and evaluate the performance of different product categories.

1	Category Performance	
2	Row Labels	Sum of Quantity
3	Electronics	150097
4	Home & Kitchen	60805
5	Sports & Outdoors	29872
6	Clothing	29638
7	Toys & Games	18215
8	Books	11513
9	Grand Total	300140
10		
11		
12	Sales Report	
13	Row Labels	Sum of Total Amount
14	2020	\$1,85,29,864.02
15	2021	\$1,82,48,574.81
16	2022	\$1,83,67,248.41
17	2023	\$1,85,13,912.19
18	2024	\$1,81,66,048.49
19	Grand Total	\$9,18,25,647.92
20		
21		
22	Best Performing Products	
23	Row Labels	Sum of Quantity
24	LED Desk Lamp	6344
25	Water Bottle	6275
26	Memory Card	6240
27	Router	6202
28	Board Game	6200
29	Grand Total	31261
30		
	Under Performing Products	
	Row Labels	Sum of Quantity
	Children's Book	5619
	Fitness Band	5724
	Car Charger	5754
	Bluetooth Speaker	5782
	Graphic Tablet	5789
	Grand Total	28668

- Fact and Dimension tables were created from the Order Details table to improve data organization and enable efficient analysis. The Fact Table contains transactional information, including Order ID, Order Date, Customer ID, Product ID, Quantity, Unit Price, Discount, Tax, Shipping Cost, and Total Amount. The Dimension Table contains descriptive product information, including Product ID, Product Name, and Category. Product ID acts as the Primary Key in the Dimension Table and as a Foreign Key in the Fact Table, establishing a relationship between the two tables using a one-to-many relationship.

Queries [4] Table.Distinct(*Filtered Rows*)

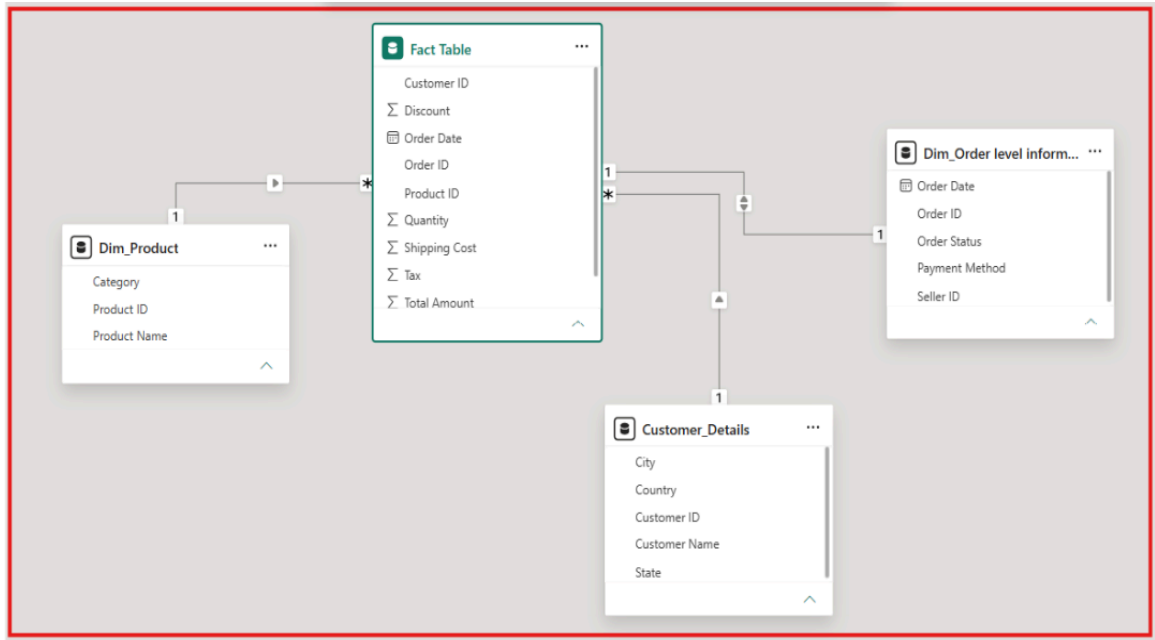
Product ID	Product Name	Category
P00003	Smartphone Case	Electronics
P00021	Backpack	Sports & Outdoors
P00011	Fitness Band	Sports & Outdoors
P00023	Cookware Set	Home & Kitchen
P00001	Wireless Earbuds	Electronics
P00034	Desk Organizer	Home & Kitchen
P00024	Yoga Mat	Sports & Outdoors
P00039	Graphic Tablet	Electronics
P00022	Water Bottle	Sports & Outdoors
P00025	Running Shoes	Sports & Outdoors
P00040	Microphone	Electronics
P00008	4K Monitor	Electronics
P00042	Projector Mini	Electronics
P00035	Desk Plant	Home & Kitchen
P00041	Webcam Full HD	Electronics
P00047	Memory Card 128GB	Electronics
P00007	Mechanical Keyboard	Electronics

Queries [4] Table.SelectColumns(Sources, {Order ID, "Order Date", "Customer ID", "Product ID", "Quantity", "Unit Price", "Discount", "Tax", "Shipping Cost", "Total Amount"})

Order ID	Order Date	Customer ID	Product ID	Quantity	Unit Price	Discount
0000025762	12-09-2023	037038270	P00003	2	5	
0000030221	25-12-2024	037009595	P00021	2	5	
0000059594	23-09-2023	037049782	P00011	4	5.01	
0000086603	23-03-2023	037042995	P00023	1	5.01	
0000010625	01-08-2022	037023333	P00001	8	5.02	
0000059279	26-12-2024	037040824	P00034	5	5.02	
0000008663	08-04-2021	037020287	P00024	3	5.03	
0000067071	15-08-2020	037023390	P00039	4	5.03	
0000008641	03-03-2020	037039988	P00011	5	5.06	
0000033789	10-07-2023	037015253	P00022	4	5.06	
0000066578	26-06-2020	037027251	P00025	2	5.06	
0000008645	26-02-2021	037040423	P00040	2	5.07	
0000003030	03-03-2022	037022287	P00025	5	5.07	

7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality.



Manage relationships

+ New relationship

⚡ Autodetect

✎ Edit

🗑 Delete

☰ Filter

☐	From: table (column) ↑	Relationship	To: table (column)	Status	
☐	Dim_Order level information (...)	1 — — 1	Fact Table (Order ID)	Active	...
☐	Fact Table (Customer ID)	* — — 1	Customer_Details (Customer ID)	Active	...
☐	Fact Table (Product ID)	* — — 1	Dim_Product (Product ID)	Active	...

- **Calculated Columns & DAX Measures:** A calculated column has been created to determine the sale amount, excluding tax and shipping costs.
- Calculated measures have been created for Total Sales, Total Orders, Total Quantity Sold, Total Customers, and Sales Growth % for 2024.

8. Analysis and Visualizations (Power BI)

Dashboard Features:

- 2 pages were created to analyse sales performance and customer behavior.

Bar Chart - Sales by Category.

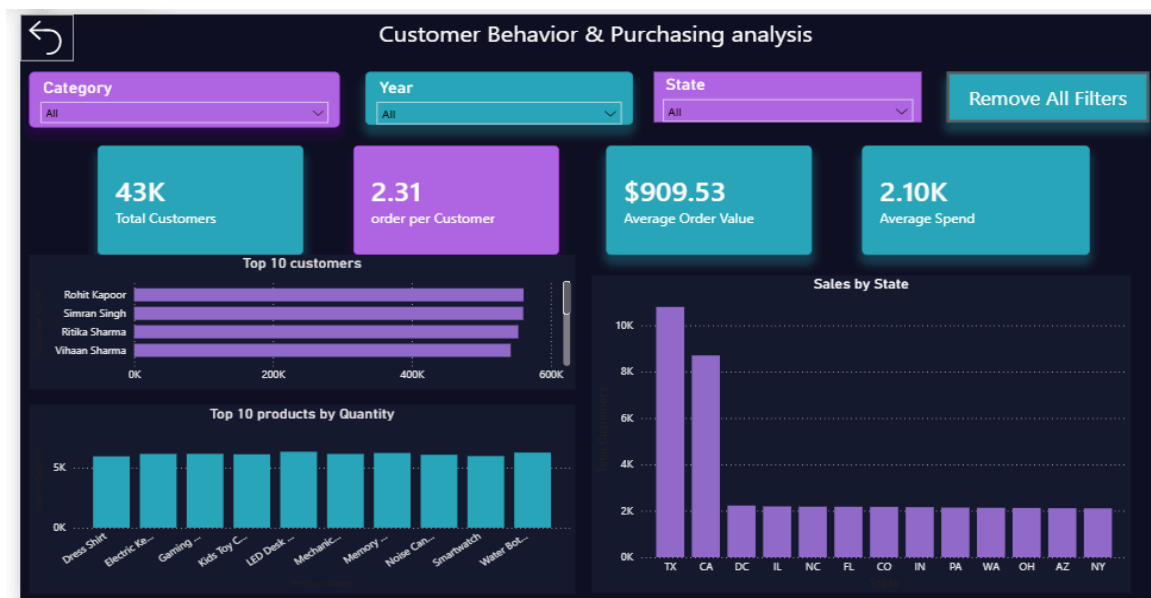
Pie Chart - Category Contribution by quantity.

Line chart - Monthly Sales Trend.

Column Chart - Top 10 products by sale, Sales by States.

- Multiple Cards were created for analysis.
- Slicers and filters were used to make the report Interactive.
- A bookmark was created to preserve the dashboard in its default state without any filters applied. A button linked to the bookmark was also added to allow users to remove all applied filters and reset the dashboard to its original state.





9. Insights & Conclusions

- **Key Findings:** Total Sales: \$90.95M | Orders: 100K | Customers: 43K
- Electronics is the top-performing category, contributing around 50% of units sold and the highest sales.
- Home & Kitchen is the second-best category and provides an opportunity for growth.
- Average Order Value: \$909.53, with 2.31 orders per customer, indicating good repeat purchasing.
- Sales Growth is -1.88%, showing a recent decline in overall performance.
- Texas and California are the strongest sales-performing states.
- Monthly sales show considerable fluctuations, indicating inconsistent sales performance.
- A small group of products and customers contributes significantly to overall sales.

Descriptive Analysis — What Happened?

The business has strong overall sales and customer activity. Electronics dominates the product portfolio, while TX and CA lead geographically. Customers make purchases more than once on average, demonstrating reasonable customer engagement.

Diagnostic Analysis — Why Did It Happen?

The negative growth and monthly fluctuations may be linked to the company's heavy dependence on Electronics and a few strong states. Lower-performing categories and states are not contributing as strongly to total sales.

Predictive Analysis — What Could Happen?

If the -1.88% growth trend continues, overall sales may remain stagnant or decline. Electronics and the leading states are likely to remain the major revenue contributors, but overdependence creates business risk.

Prescriptive Analysis — What Should Be Done?

- Protect and strengthen the Electronics category.
- Expand Home & Kitchen and other underperforming categories.
- Increase customer retention and repeat purchases.
- Use cross-selling, upselling, and product bundles to increase AOV.
- Investigate the causes of the -1.88% sales decline.
- Expand marketing efforts in lower-performing states.
- Reduce dependence on a single category by diversifying the product portfolio.

10. Conclusions

The business has a strong sales base and valuable customers, but the negative growth rate is a key warning sign. The priority should be to identify the cause of declining sales, strengthen customer retention, grow secondary categories, and expand into weaker markets to achieve sustainable growth.